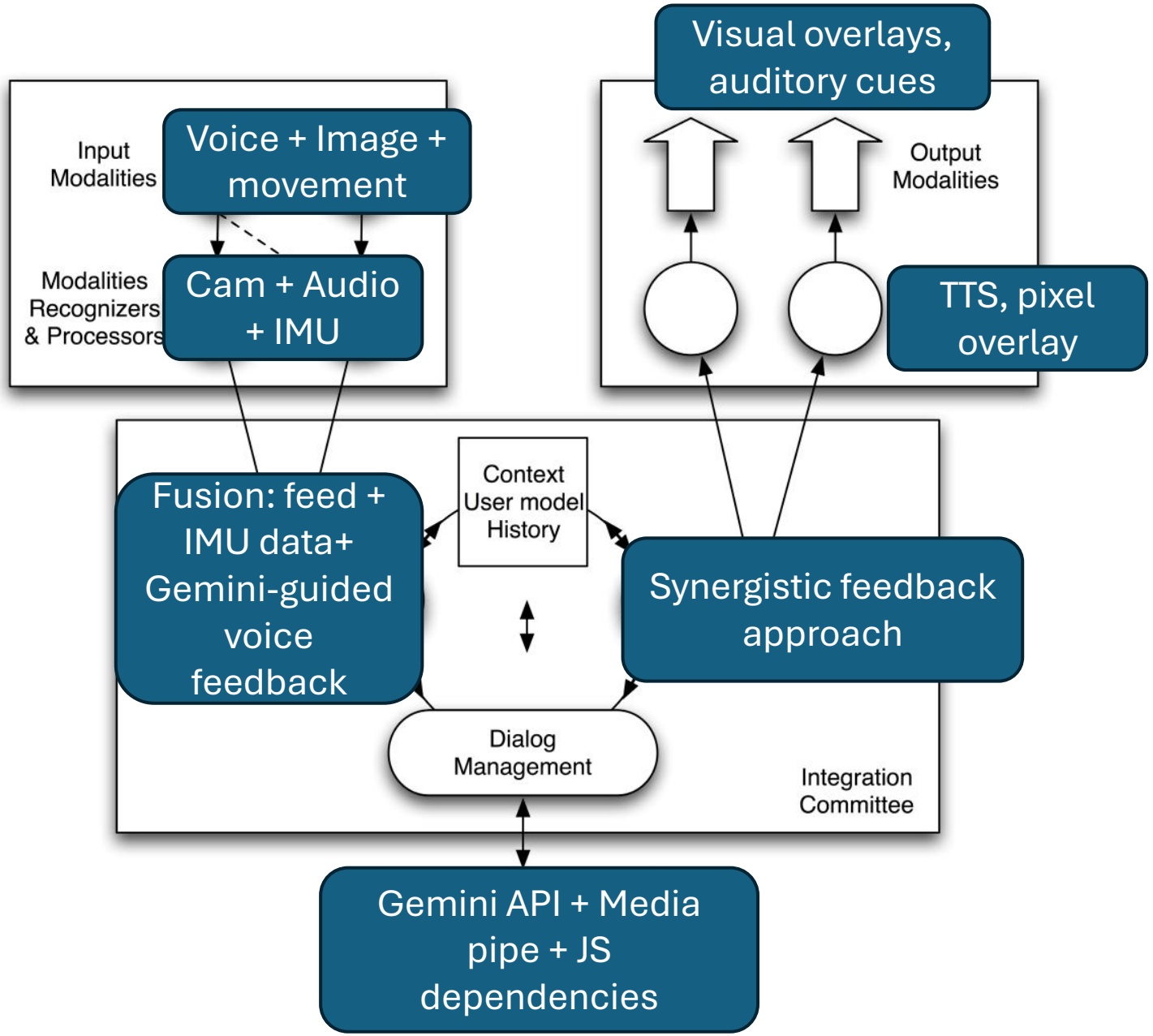
Impulse: Synergistic multimodal sports training

The problem:

- **Static training** (lack of gamification leads to low engagement)
- **Injury risk** (absence of RT, physio-informed posture correction)
- **Data saturation** (overwhelming raw metrics that act as noise more than actionable feedback)
- **Feedback void** (Inaccurate proprioceptive feedback and lack of post-execution quality feedback)



Architecture & phased implementation plan (14 weeks)

Input modalities fusion (Python): Optical recognition (video) + Hardware kinematics (IMU sensors) into a unified model.
Dialog management & context: Fusion of input (Python + HTML/JS), the dialog management oversees the Gemini API as an LLM-driven dialog manager to add dynamism to the feedback.

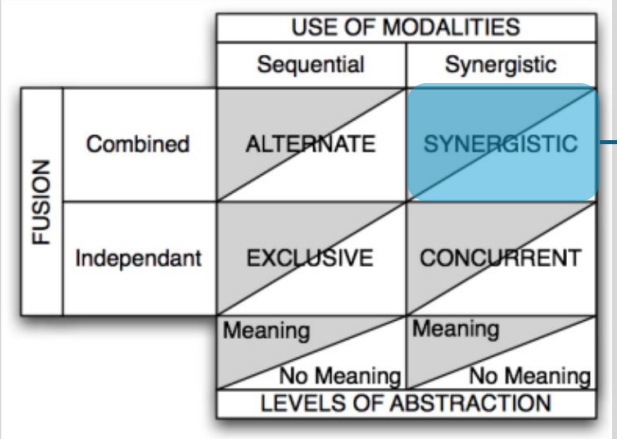
Output modalities fission (Python): Translating LLM's payload and mediapipe's inferences into a gamified visual and metrics with clear acoustic coaching cues.

We are combining multiple sensory information into an LLM-driven manager that provides insightful feedback.

CASE/CARE Models

CASE – "*how*" modalities are combined

CARE – "why" modalities are combined



Nigay and Coutaz 1993

Sprint 1 (visual Foundation) W4-W6
- Python integration of camera feed, audio processing and basic CV recognition.

Sprint 2 (Cognitive engine) W7-W9
- Integration of the Gemini API and GUI essentials

Sprint 3 (fusión + HW) W10-W14
- Wearable integration of IMU; polish of LLM inferences; potential integration of other systems*

*potential systems to add: V-JEPA for physics informed inferences, Robust detections of IMU outside of angular velocity w. PINNs, Integration in a Flutter-based web App